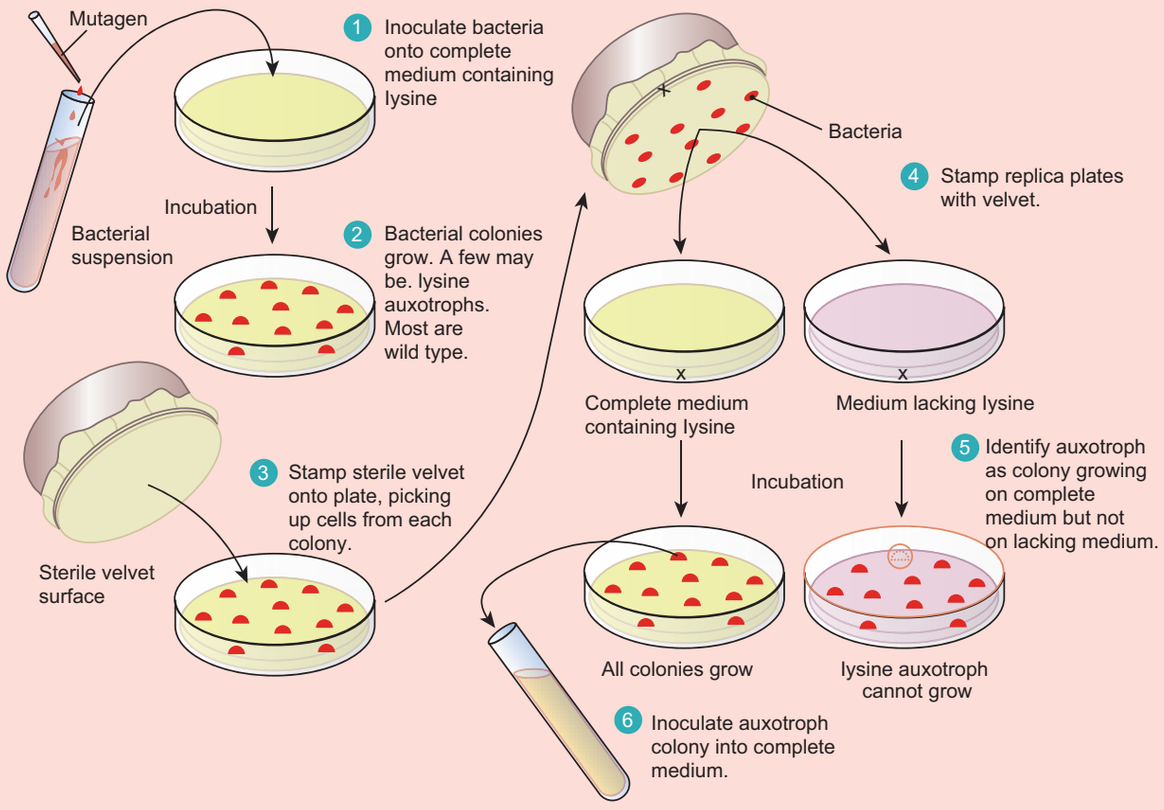
Infobits

Isolation and detection of Mutants

Once mutations are induced, then, they must be detected if they are to be studied. Selection and screening procedures historically have helped geneticists isolate mutants of interest from a heterogenous mixture in a mutagenized population. When isolating mutants of a particular organism, one must know the normal or wild type characteristics so as to recognize an altered phenotype. A suitable detection system for the mutant phenotype under study also is needed. Detection systems in bacteria and other haploid organisms are straightforward because any new allele should be seen immediately, even if it is recessive mutation. The detection of mutants can be direct and complex. For example, the replica plating technique is used to detect auxotrophic mutants (mutants which are deficient in synthesizing a particular biochemical compound). **Replica plating technique** distinguishes between mutant and wild type strain based on their ability to grow in the absence of a particular biosynthetic end product Figure below. A lysine auxotroph, for instance, will grow on lysine supplemented media but not on a medium lacking an adequate supply of lysine because it cannot synthesize this amino acid.



TN_GOVT_XII_Micro_Biology_CH12.indd 262 04-03-2019 10:51:47